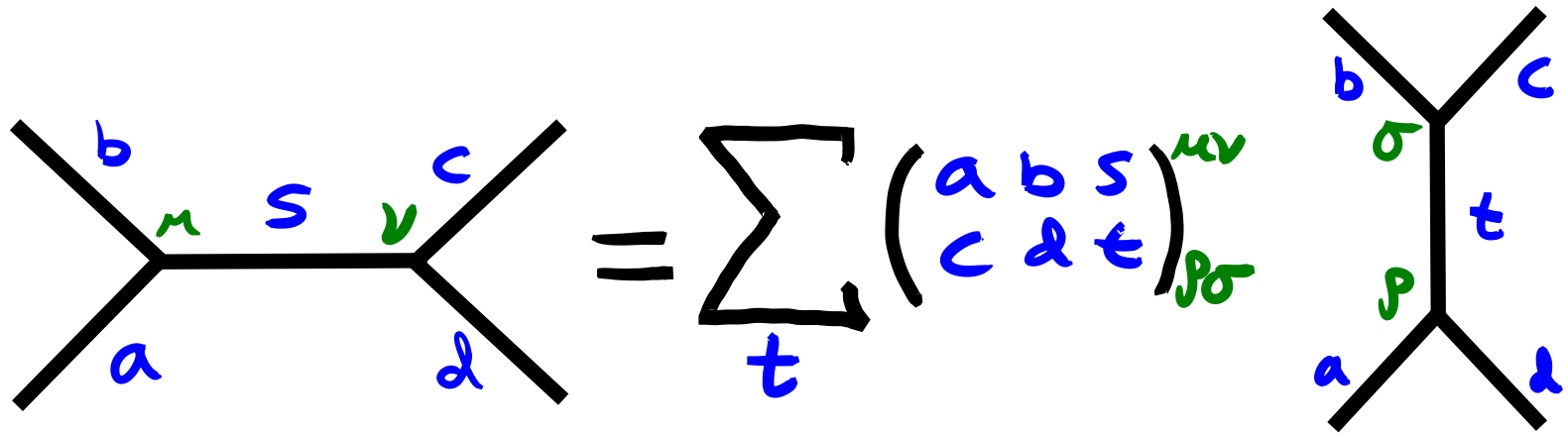
Graph Games From Rep. Theory

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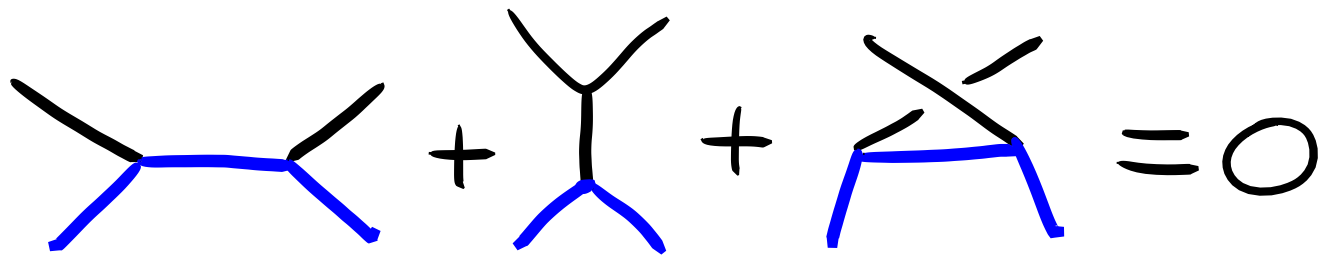
Motivation

Gauge thy Feynman expansion

$$\mathcal{A}_n(1^a 2^b \cdots n^\gamma) = \sum_{\text{graphs } \Gamma} \overbrace{C_\Gamma^{ab\cdots n}}^{\text{Rep. thy tensor}} \underbrace{a^\Gamma}_{\text{kinematics}}$$

C_Γ not indep.!

eg



Motivation

Gauge thy ~~Feynman Expansion~~

$$\mathcal{A}_n(1^a 2^b \cdots n^n) = \sum_i \underbrace{c_i^{ab\cdots n}}_{\text{Indep. tensor basis}} \overbrace{A^i}^{\text{Partial amp.}}$$

c_i Indep. $\Rightarrow$ A^i gauge inv.!

$\mathcal{A}_n$ à la Feynman **NOT** gauge inv.

Tensor Bases

By a basis $\{\mathcal{C}_i\}$, we mean:

- Minimal & Spanning ,
- Orthogonal ,
- Valid for all orders in P.T.,

n.b. | DDM, Traces, etc do NOT Satisfy this definition!

For irreps $a, b, \cdots, n$ of Simple Lie $\mathfrak{g}$, a basis of $\{\mathcal{C}_i\}$ is:

$$\mathcal{C}_i^{ab\cdots n} = \quad \text{(a, b, c, } \mu, \nu, I_1, \ldots, n-2, \rho, \sigma, I_{-1}, n-1, n\text{)} \quad \Big| \quad \text{(a, b, c, } \mu\text{)} \quad := \quad \text{"3-j" or "Clebsch Gordan"}$$

Questions We asked (& answered!)

Q For given g & $a, b, \cdots, n$, how many c_i?

Cool integer sequences!
Asymptotics!
Recurrences!

2412.21189
J. Boursaily
M. Plesser
C. Versu

Q How to construct c_i?
Do c_i have interesting properties?

Yes! Yang-Baxter,
Pentagon & Hexagon,
etc...

25XX.XXXX
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M. Plesser
C. Versu

Today

Graphical Games

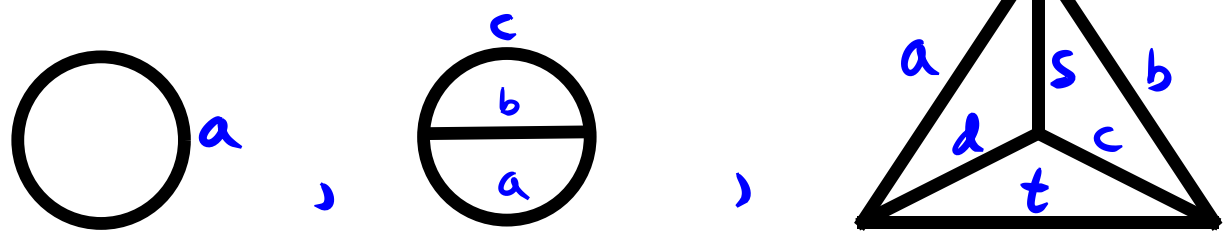
C.f. Yutsis, Penrose, Cvitanović, etc...

Given tensors C_j, in the $\mathcal{C}_i$ basis:

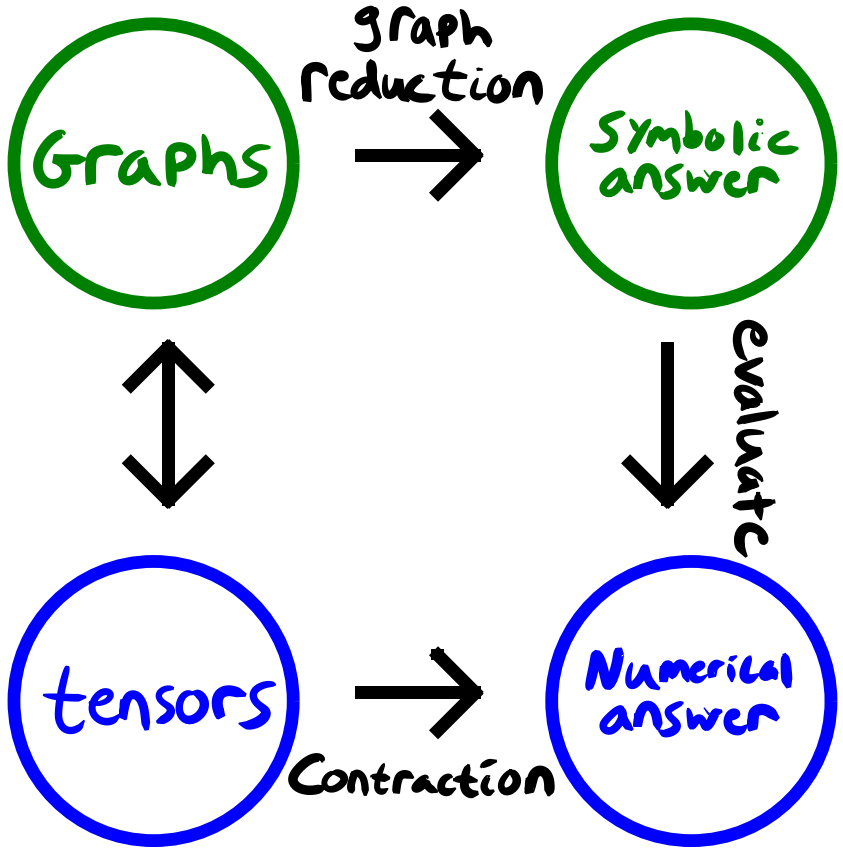
$$C_j^{ab\cdots n} = \sum_i \mathcal{C}_i^{ab\cdots n} \frac{\left(\mathcal{C}_{i\,a'b'\cdots n'} C_j^{a'b'\cdots n'}\right)}{\left(\mathcal{C}_{i\,a''b''\cdots n''} \mathcal{C}_i^{a''b''\cdots n''}\right)}$$

Correspond to Vacuum graphs

eg



Evaluation



1 Tensor Contraction

+) Straight forward

−) Computationally Costly

2 Graph Reduction

+) Scales well

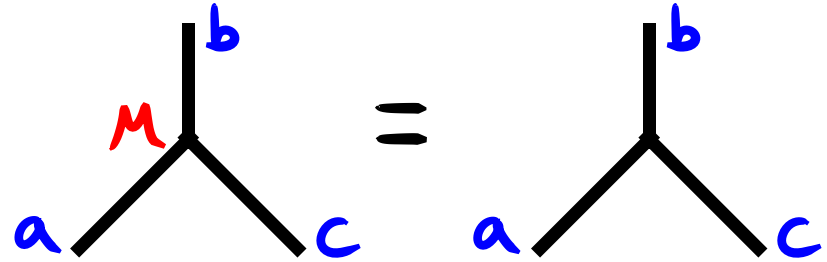
−) Detailed bookkeeping

Let's explore option 2!

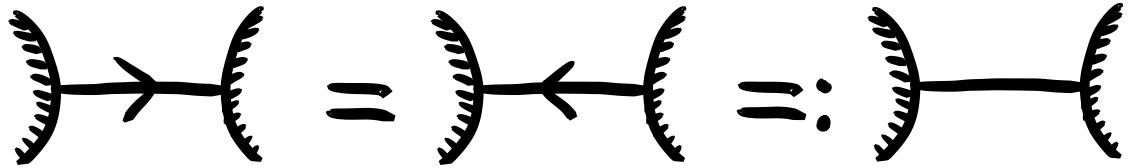
Setup

Take $\boxed{9} = SU(2)$

Why? 1 No vertex Multiplicity!



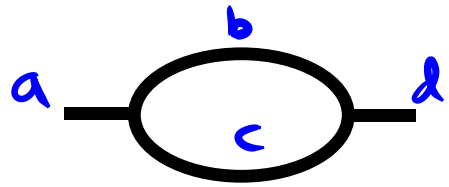
2 We can use Undirected graphs*!



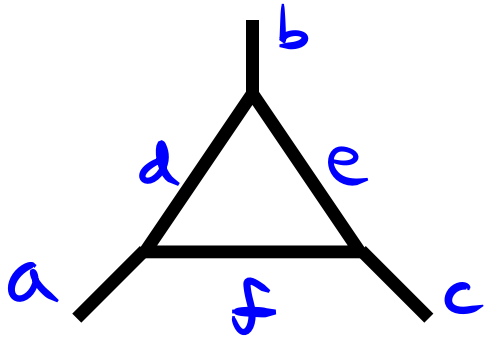
3 SU(2) 6Js known analytically!

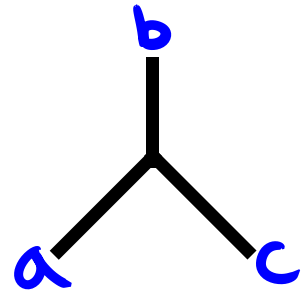
Consider this a Toymodel for general $\boxed{9}$

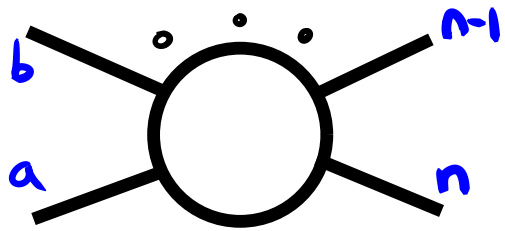
Rules of the game

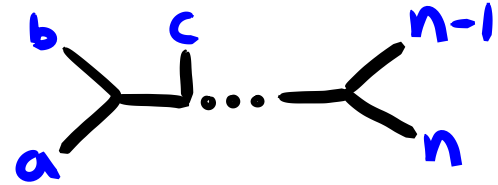
① 

$= \delta_{ad} \begin{pmatrix} a & b & c \\ b & a & 0 \end{pmatrix} d_b$ 

② 

$= \begin{pmatrix} f & a & d \\ b & e & c \end{pmatrix} \begin{pmatrix} c & e & f \\ e & c & 0 \end{pmatrix} d_e$ 

③ 

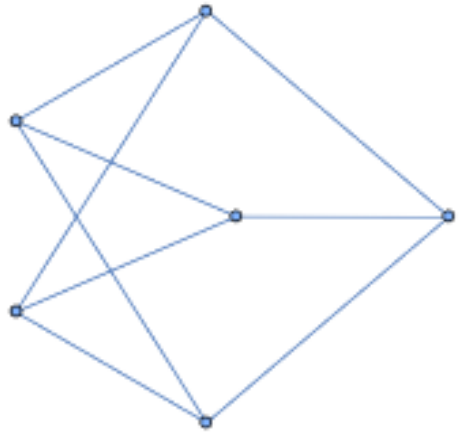
$= \sum (\quad) (\quad) \cdots \times$ 

Using ①, ②, ③ : graph $\mapsto \sum \prod \begin{pmatrix} i & j & k \\ l & m & n \end{pmatrix}$

Results

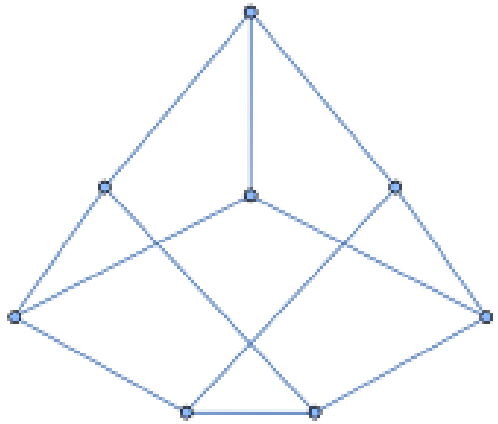
Some Classic examples

9J



$$\text{sum}\Big[(-1)^{i_1+2j_1+2j_2+j_3+j_4+j_5+j_6+j_7+j_8+j_9}\begin{pmatrix} j_1 & i_1 & j_2 \\ i_1 & j_1 & 0 \end{pmatrix}\begin{pmatrix} j_1 & j_4 & j_7 \\ j_8 & j_2 & i_1 \end{pmatrix}$$
$$\begin{pmatrix} j_1 & j_9 & j_5 \\ j_9 & j_1 & 0 \end{pmatrix}\begin{pmatrix} j_8 & i_1 & j_4 \\ j_6 & j_3 & j_9 \end{pmatrix}\begin{pmatrix} j_9 & i_1 & j_6 \\ j_2 & j_5 & j_1 \end{pmatrix}\begin{pmatrix} j_9 & j_8 & j_3 \\ j_8 & j_9 & 0 \end{pmatrix} d_{i_1} d_{j_1} d_{j_8} d_{j_9},$$
$$\{i_1, \text{Max}[|j_1 - j_2|, |j_4 - j_8|], \text{Min}[j_1 + j_2, j_4 + j_8]\}\Big]$$

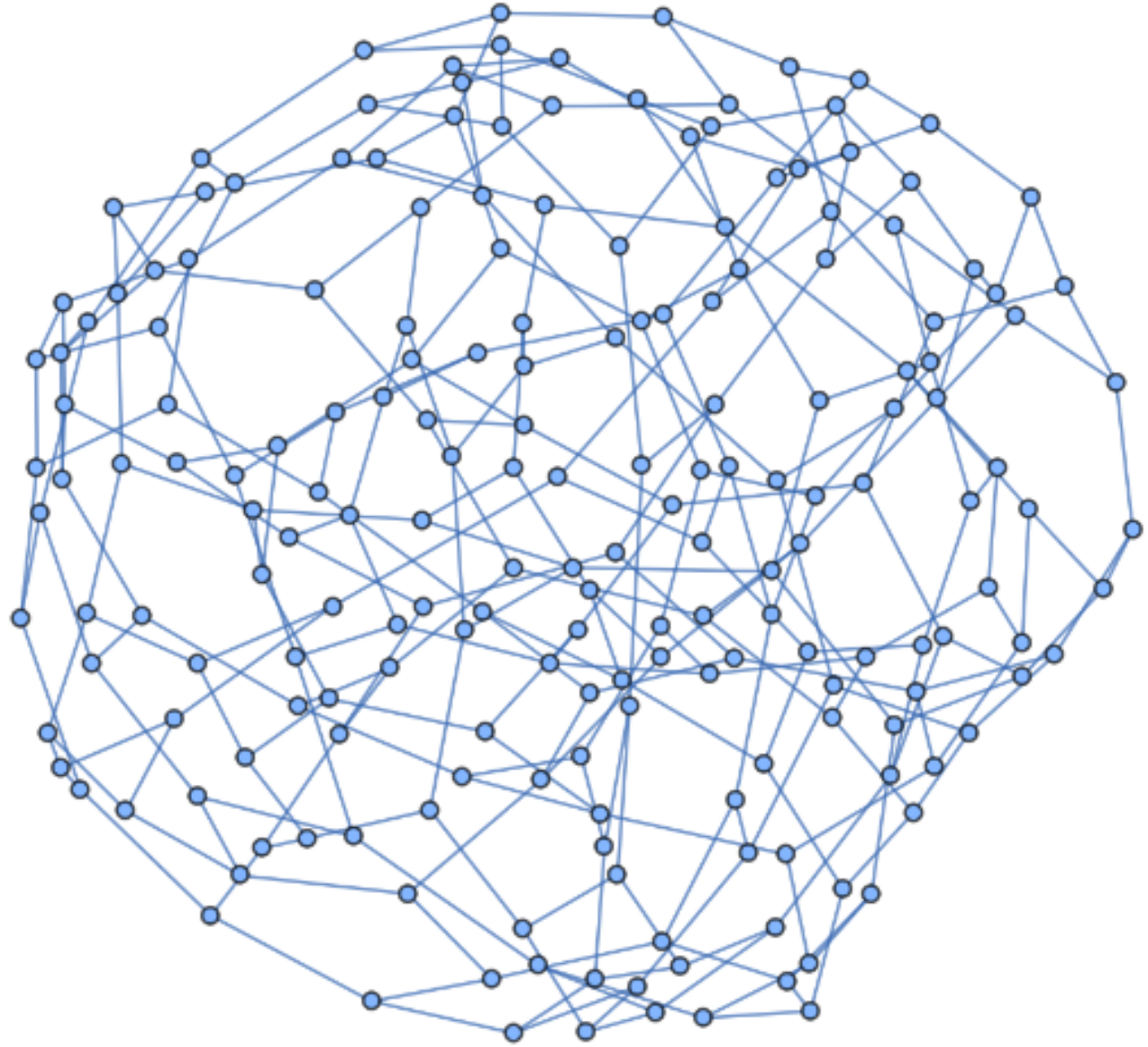
12J



$$\text{sum}\Big[(-1)^{2i_1+i_2+2j_1+2j_2+j_3+j_4+j_5+2j_6+j_7+j_9+2j_{10}+j_{11}+j_{12}}\begin{pmatrix} i_1 & i_2 & j_1 \\ i_2 & i_1 & 0 \end{pmatrix}$$
$$\begin{pmatrix} i_1 & j_4 & j_{10} \\ j_{11} & j_1 & i_2 \end{pmatrix}\begin{pmatrix} i_1 & j_{12} & j_6 \\ j_{12} & i_1 & 0 \end{pmatrix}\begin{pmatrix} j_4 & j_8 & j_9 \\ j_8 & j_4 & 0 \end{pmatrix}\begin{pmatrix} j_6 & j_2 & j_5 \\ j_8 & j_{12} & i_1 \end{pmatrix}\begin{pmatrix} j_8 & i_1 & j_2 \\ j_{10} & j_9 & j_4 \end{pmatrix}$$
$$\begin{pmatrix} j_{11} & i_2 & j_4 \\ j_7 & j_3 & j_{12} \end{pmatrix}\begin{pmatrix} j_{12} & i_2 & j_7 \\ j_1 & j_6 & i_1 \end{pmatrix}\begin{pmatrix} j_{12} & j_{11} & j_3 \\ j_{11} & j_{12} & 0 \end{pmatrix} d_{i_1} d_{i_2} d_{j_8} d_{j_{11}} d_{j_{12}},$$
$$\{i_1, \text{Max}[|j_2 - j_8|, |j_6 - j_{12}|], \text{Min}[j_2 + j_8, j_6 + j_{12}]\},$$
$$\{i_2, \text{Max}[|i_1 - j_1|, |j_4 - j_{11}|], \text{Min}[i_1 + j_1, j_4 + j_{11}]\}\Big]$$

Results

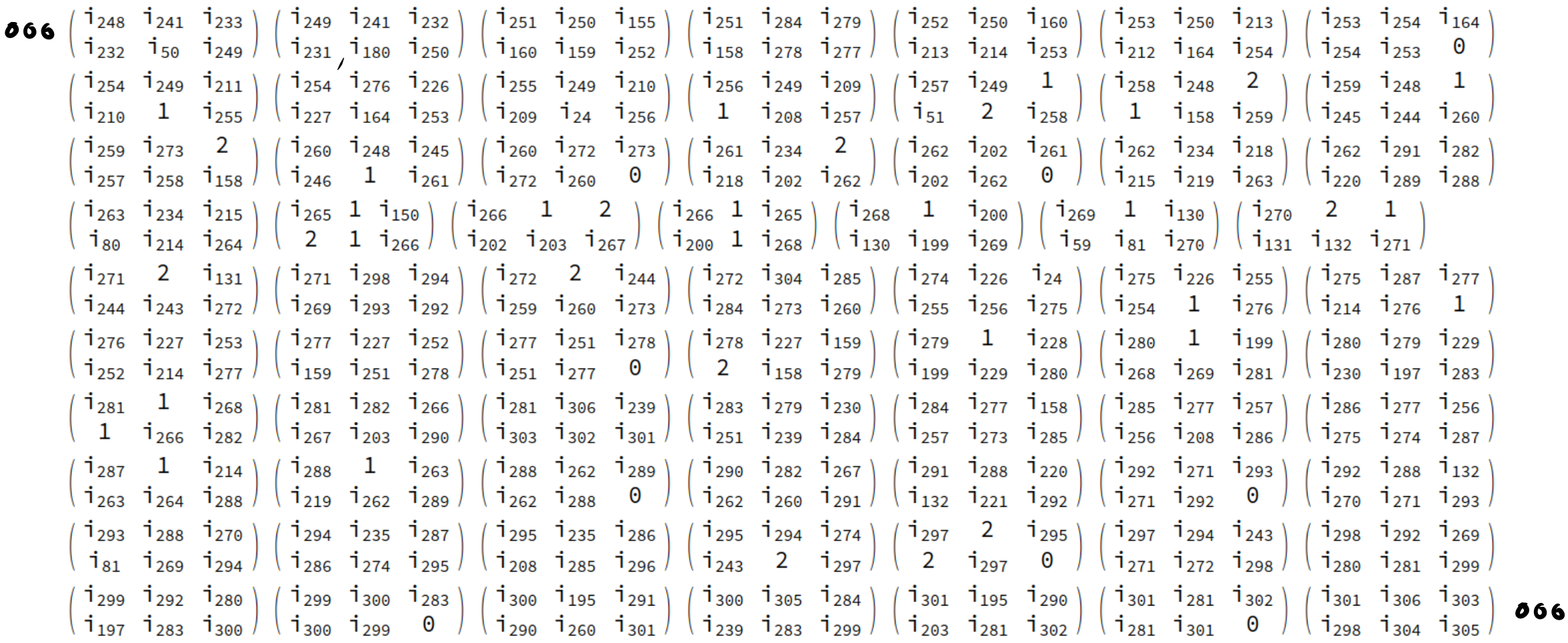
How about a 196J?



Results



Results



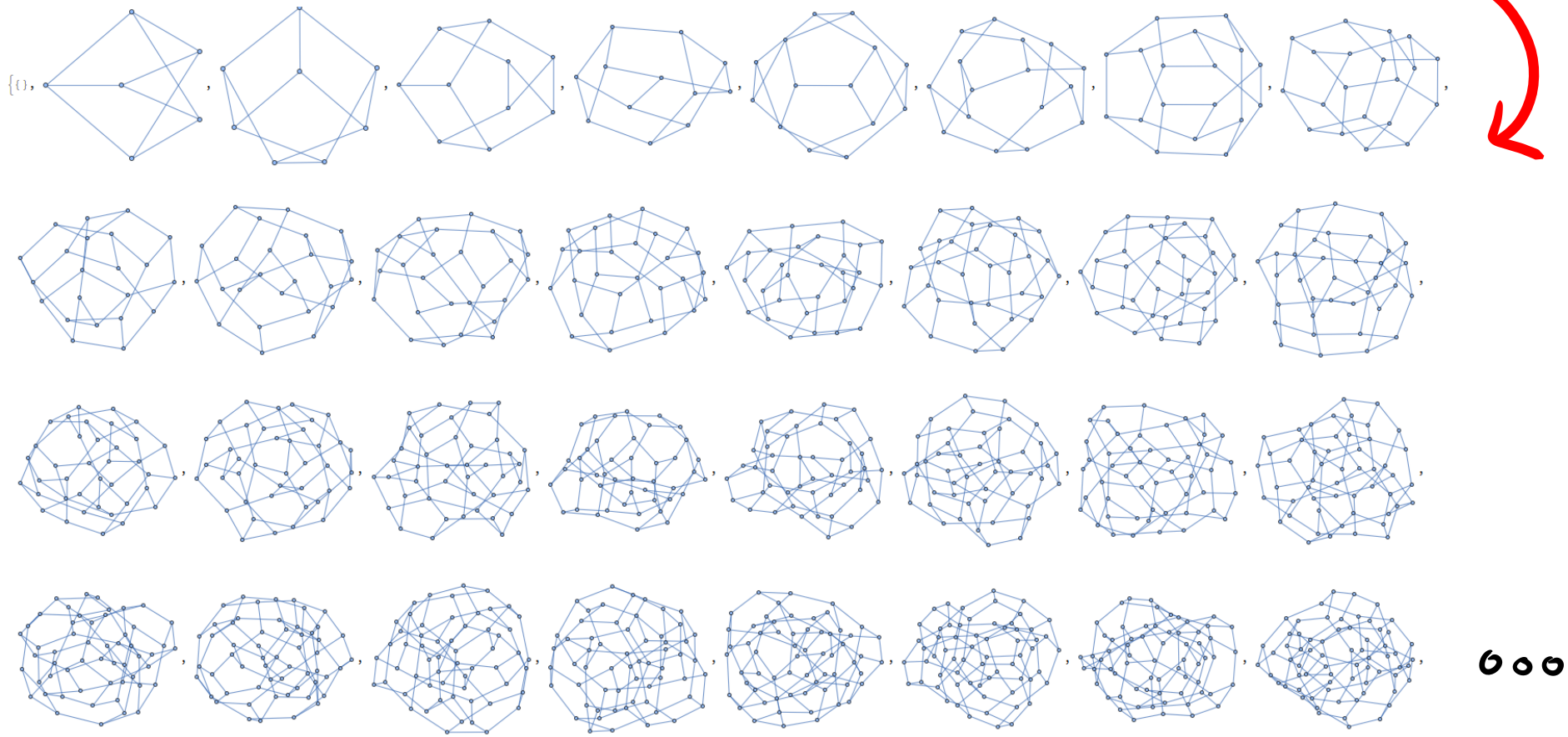
That's a lot of 6Js!

Results

$N_e = 196$ $N_v = 294$

Reduction time ~ 75 s

Also learned all these! Recursively!



600

Thanks

Future?

- General 9
- Minimal sets of 6Js
- Generalized Biedenharn-Elliot
- etc...

Thanks